Edit Distance 40

Given two strings word1 and word2, find the minimum number of operations required to convert word1 into word2.

You process word1 using a cursor from left to right. At your current position, you may perform one of the following operations:

• Match: If the current character of word1 matches the required character in word2, move past both characters (cost 0).

• Replace: Replace the current character of word1 to match the required character in word2, and move past both characters (cost 1).

• Delete: Delete the current character of word1 (cost 1). Side effect: All remaining unprocessed characters to the right of your cursor immediately undergo ROT13.

• Insert: Insert the required character for word2 at your current position (cost 1). Side effect: All remaining unprocessed characters to the right of your cursor immediately undergo ROT13.

ROT13 shifts a lowercase letter forward by 13 positions in the alphabet, wrapping around (e.g., 'a' becomes 'n', 'm' becomes 'z', 'n' becomes 'a'). Applying ROT13 twice restores the original character.

Input Format

The first line contains a string, word1, denoting the first word consists of lowercase English letters.

The second line contains a string, word2, denoting the second word consists of lowercase English letters.

Constraints

1 <= |word1| <= 500

1 <= |word2| <= 500

Output Format

The output contains an integer value

Sample Input 0

```
am
z
```

Sample Output 0

```
1
```

Explanation 0

We need to transform am into z. We delete the first character a for a cost of 1. Because of the deletion, all remaining unprocessed characters in the string undergo ROT13. The remaining character m shifts 13 places to become z. Now, our current string is z, which matches the target z perfectly for a cost of 0. The total minimum cost is 1.

Sample Input 1

```
ab
cd
```

Sample Output 1

```
2
```

Explanation 1

We need to transform ab into cd. No single insertion or deletion creates a beneficial ROT13 cascade for these specific characters. The optimal strategy is to replace a with c (cost 1) and b with d (cost 1). Replacements do not trigger any ROT13 shifts. The total minimum cost is 2.

Sample Input 2

```
abc
def
```

Sample Output 2

```
3
```

Explanation 2

We need to transform abc into def. Similar to the previous case, no sequence of insertions or deletions creates a beneficial cascade to match these characters. The total minimum cost is 3. We simply replace a with d, b with e, and c with f. Each replacement costs 1.